

Statistics and Scientific Methods

University of Oxford

PRACTICAL 1. Scientific Method: Introduction to practical statistics using R



- Introduction to the practical side of the course
- Checking R: Updating R/R Studio and installing packages
- Reproducible science

Hilary Term: Statistical analysis using R

Week	Topic	Format
1	Scientific method R practical (data entry etc.)	Lecture online Practical
2	Statistical inference R practical	Lecture online Practical
3	Probability theory R practical	Lecture online Practical
4	Hypothesis testing R practical	Lecture online Practical
5	Analysing proportions R practical	Lecture online Practical
6	Contingency analysis Chi-squared test of association: Mendelian genetics	Lecture online Practical
7	Origin of SARS-COV-2	Practical
8	Drop-in question session	Drop-in session

Checklist (Do/Have I...?)

- ✓ Remember Andy Hector's lectures on R and R Studio
- ✓ Watch the lecture recording on "Statistics and Scientific Methods" by Jeri Wrigh
- ✓ Assignment: Chapters 1 and 2 Whitlock, M.C. and Schluter, D. 2020. Analysis of Biological Data. 3rd ed. W.H. Freeman and Co.: New York, NY.
- ✓ Assignment: Chapter 1 and 2
Macmillan Achieve: Homework Problems and Assignment Problems
- ✓ Own a 'free-standing' calculator



THE ANALYSIS OF
BIOLOGICAL DATA
WHITLOCK AND SCHLUTER
THIRD EDITION

Achieve
macmillan learning

Download the .Qmd file from Canvas and complete the exercises

And contrast your answers with us, we are here to help!